How Open Source Tools Power Modern IT Operations

Open source tools have not replaced enterprise IT platforms; they have become the connective layer that makes modern operations possible.

Historically, enterprise IT infrastructure operations were characterised by predictability. Servers were provisioned, applications were deployed, and most monitoring consisted of ensuring that systems were 'up'. The tools used to perform these activities tended to be proprietary, tightly coupled with specific platform environments and engineered for static environments that rarely changed.

Now, this model has changed dramatically.

The operations supporting modern IT infrastructures can be highly decentralised systems, hybrid and multi-cloud deployments/platforms, containerised applications, and continual delivery pipelines. Visibility, reliability, and adaptability are no longer an option but, rather, core components of IT operations. Quietly but surely, open source technologies are providing the foundations to enable enterprises to meet these demands well.

This evolution/process did not occur overnight; it was born out of the real-world need to operate environments that included a combination of Windows and other OSes, where traditional tools were incapable of addressing the increasing complexity of operations.

In its early years, open source was used mainly to assist IT staff with basic system monitoring and diagnostics. Open source tools were useful because they addressed specific issues such as improving system health, automating tasks, or collecting logs. However, these tools typically did not help to define the strategy to support these IT needs.

Most of the time, IT data came from proprietary vendors. IT departments relied on IT monitoring tools to monitor individual servers, and alerts from those tools were primarily based on pre-defined performance thresholds. Investigating and diagnosing the root cause was primarily performed manually by the IT staff members, who knew how the system's components worked together.

As systems became more complex through virtualisation and centralisation, and applications began to spread across different locations, it became increasingly difficult to correctly diagnose any outage or failure on any component of an IT system. Instead of being able to verify if the server was 'on', IT staff were now required to determine 'why' the service was not functioning correctly.

Addressing operational complexity with open source

Cloud architecture, microservices, and distributed workloads created increased operational complexity, which caused companies to move away from using static dashboards in conjunction with siloed applications. Due to the ability of